

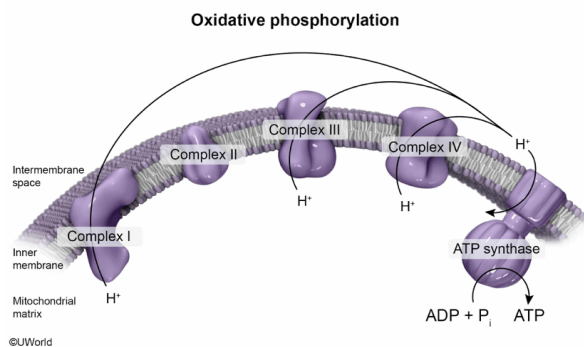
The electron transport chain results in the generation of:

- ✔ ☒ A. a proton concentration gradient across the inner mitochondrial membrane, which drives phosphorylation of ADP. [84%]
- ☐ B. a proton concentration gradient across the inner mitochondrial membrane, which reduces NAD^+ in a coupled reaction. [5%]
- ☐ C. an electron concentration gradient across the inner mitochondrial membrane, which drives phosphorylation of ADP. [4%]
- ☐ D. an electron concentration gradient across the inner mitochondrial membrane, which drives oxidation of NADH in a coupled reaction. [4%]

Correct

85% answered correctly

Explanation:



Oxidative phosphorylation is the final step in [cellular respiration](#), taking place within the mitochondria. It is driven by energy from NADH and FADH_2 , which are produced during glycolysis and the citric acid cycle. These molecules enter the **electron transport chain**, where they initiate the transfer of **electrons** through a series of energetically favorable [redox reactions](#) that end with O_2 as the final electron acceptor. These coupled redox reactions occur in [transmembrane complexes](#) that reside on the inner mitochondrial membrane between the **mitochondrial matrix** and the **intermembrane space**.

Because these reactions are energetically favorable, the excess energy can be used to pump **protons (H^+)** from the mitochondrial matrix into the intermembrane space. This establishes both a **proton concentration gradient** and a charge gradient (collectively called an [electrochemical gradient](#)) across the membrane. The gradient drives the movement of protons from the intermembrane space (higher concentration) back into the matrix (lower concentration) through the [ATP synthase](#) complex. This **energetically favorable movement of H^+** down its electrochemical gradient (ie, chemiosmosis) provides ATP synthase with the **energy needed to phosphorylate ADP**, forming ATP.

(Choice B) Oxidative phosphorylation involves the **oxidation** of NADH, not the reduction of NAD^+ . In addition, redox reactions are driven by the transfer of *electrons*, not by a proton concentration gradient.

(Choices C and D) The electron transport chain does not *generate* an electron gradient across the membrane but *transfers* electrons from one chemical species to another in successive coupling reactions. This process provides the energy needed to pump protons from the mitochondrial matrix to establish an electrochemical gradient.

Educational objective:

In oxidative phosphorylation, a series of energetically favored redox reactions provides the energy to pump protons from the mitochondrial matrix into the intermembrane space, creating an electrochemical gradient that drives ATP synthesis.

Time spent:0 sec

Subject:
Biochemistry

QID:402766

System:
Metabolic Reactions